The Controversy over "Chemical Restraint" in Acute Care Psychiatry

GLENN W. CURRIER, MD, MPH

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In many healthcare settings, medications are considered a less invasive alternative to the use of physical restraint for agitated patients experiencing a behavioral crisis, a practice that is often referred to as "chemical restraint." However, recent federal regulations appear to equate chemical and physical restraint and to characterize both as extraordinary practices that should not be undertaken lightly. Although many clinicians consider the term "chemical restraint" pejorative, since it does not reflect the possibility that forced medication may be clinically necessary and have a beneficial effect, the term is embedded in recent regulatory language. The author first reviews the controversy over the concept of chemical restraint as it has developed in the mental health literature and regulatory policy. As yet there is no consensus among clinicians or policy makers whether such use of medications is a form of coercion or a form of patient-focused intensive care. The author then discusses precipitants of emergency care and clinical factors and situations that may lead to the use of medications in a way that might be considered chemical restraint. Such factors include clinical and demographic characteristics of patients, institutional characteristics, and staff perception and attitudes. In the final section of the article, the author reviews the recommendations concerning the emergency use of medications given in the *Expert Consensus Guidelines on the Treatment of Behavioral Emergencies* and discusses treatment developments that have occurred since the time of the survey on which those guidelines were based. (*Journal of Psychiatric Practice* **2003;9:59–70**)

KEY WORDS: restraint, chemical restraint, seclusion, involuntary medication, forced medication, regulatory policy, expert consensus guidelines, behavioral emergency

The use of physical or "chemical" restraint and seclusion in institutional and community settings remains an area of controversy, which has been compounded by a lack of systematic data collection or careful evaluation of indications, methods, outcomes, and adverse events.[1–9] The deleterious physical and psychological effects of restraint and seclusion are increasingly being recognized and the use of such interventions has received tremendous scrutiny from regulatory agencies and the popular press. At the same time, the changing nature of psychiatric care has made the management of agitated and aggressive patients more challenging. While the number of patients requiring psychiatric care continues to increase, the number of inpatient beds per capita has steadily decreased.[10] State laws allowing involuntary commitment of mentally ill patients are predicated on dangerousness. As a result, potentially violent people of all ages are increasingly being funneled away from longer-term psychiatric settings (e.g., state hospitals) to emergency departments, jails, prisons, shelters, and community-based residential settings. In addition, the "sickness threshold" for access to limited psychiatric hospital beds has increased, resulting in a higher concentration of seriously disturbed and potentially violent patients in psychiatric facilities.

In many healthcare settings, medications are considered a less invasive alternative to physical restraint, a practice commonly referred to as "chemical restraint." However, in recent federal policy, chemical and physical restraint are generally treated in the same way.

For many, the term "chemical restraint" is pejorative, with inherently negative implications. Such a term does not fully reflect the possibility that forced medications were clinically necessary and had a beneficial effect.

CURRIER: University of Rochester School of Medicine.

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Please send correspondence and reprint requests to: Glenn W. Currier, MD, MPH, University of Rochester Medical Center, 601 Elmwood Ave, Rochester, NY 14642.

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Alternative terms such as "rapid tranquilization" have appeared, although no nomenclature is widely accepted. Since "chemical restraint" is embedded in most of the recent regulatory language, this term is used throughout this article. Nonetheless, it seems clear that "chemical restraint" and "involuntary treatment" are inter-related but not equivalent concepts. In this article, I first review the controversy over the concept of chemical restraint as it has developed in the mental health literature and regulatory policy and consider when, if ever, the use of medications might be considered chemical restraint. I then consider clinical factors and situations that may lead to the use of medications in a way that might be considered chemical restraint.

In the last section of this article, I review the recommendations concerning use of emergency medications (whether considered chemical restraint or not) derived from the *Expert Consensus Guidelines on the Treatment of Behavioral Emergencies*[11] and discuss treatment developments since the survey on which the guidelines were based was conducted in 2000. This survey of experts was undertaken because the literature to guide clinicians in best practices for behavioral emergencies is very limited The *Guidelines* address indications, contraindications, and recommended techniques for treating behavioral emergencies across the age spectrum. This work represents an important initial attempt to clarify both current and recommended practices for the use of physical and chemical restraint and seclusion. Nonetheless, basic issues remain to be addressed, including an accepted definition of a behavioral emergency, optimal techniques for intervention, appropriate training tools and standards for clinicians and staff involved in applying restraints and seclusion, and defining necessary oversight or monitoring methods to track the uses, benefits, and adverse effects of such measures.

THE CONTROVERSY OVER CHEMICAL RESTRAINT VERSUS TREATMENT

Overview of the Controversy

Articulating appropriate use of involuntary psychotropic medication is made difficult by several unresolved questions.

- Is forced medication ever appropriate?
- If so, what are reasonable thresholds for such intervention?
- What differentiates "chemical restraint" from "acceptable" forced medication?
- What defines "voluntary" versus "involuntary" care?
- Is chemical restraint more or less intrusive than physical restraint?

Clinicians and policymakers disagree on whether medications given urgently to ameliorate aggression or agitation constitute treatment or a form of restraint. Aggression, per se, is not considered a mental disorder, although it constitutes a diagnostic anchor point for several major mental illnesses. Antipsychotic medications of various classes, a mainstay of emergency treatment, have effects on a variety of neuroreceptors. Because these agents require days to weeks to exhibit effects on the positive symptoms of psychosis, clinicians in essence make use of the extensive side-effect profiles of these agents to achieve rapid sedation without immediately affecting the underlying pathology. This concept is intermingled with the unfortunate theory of "rapid neuroleptization" that was prominent in the 1960s and 70s.[12] High doses of conventional antipsychotics, most commonly haloperidol, were given over a short time frame in order to "lyse" positive symptoms. Although clinicians may have seen decrements in untoward or aggressive behavior, this effect may have been more related to drug-induced parkinsonism than to any effective treatment. Such experiences were unpleasant for patients and contributed to a public perception that undue and extreme coerciveness were inherent in psychiatric care.

Related to the ambiguity involved in defining treatment versus restraint, it is also unclear in some cases if treatment is truly voluntary or covertly involuntary. If an agitated patient is given a sedative intramuscularly while in restraints, it is reasonable to conclude that this is involuntary treatment. However, the distinction is less clear if an agitated patient takes oral medication under implied or stated duress.

At the extreme end of this controversy, Crumley asserted that the term chemical restraint should never be applied to any p.r.n. psychotropic medication given after an aggressive or self-injurious incident. He believed that psychotropic agents were designed to promote sedation by decreasing anxiety or panic, restoring moods to normal, and improving cognitive abilities by restoring reality testing and, as such, should be regarded as a form of treatment rather than restraint.[13]

Definitions of Chemical Restraint

A chemical restraint is defined by the Center for Medicare and Medicaid Services (CMS) as "any drug

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used to restrict a patient's freedom of movement or for emergency control of behavior, but is not a standard treatment for his or her medical or psychiatric condition."[14]

The Joint Commission on Accreditation of Healthcare Organization defines chemical restraint as "The inappropriate use of a sedating psychotropic drug to manage or control behavior."[15]

These definitions lack precision. The CMS definition, while linking the application of treatment to the presence of a condition, fails to specify if a "disease entity" is necessary or if more protean symptoms suffice. Even champions of the DSM classification scheme acknowledge the lack of precision and inherent fluidity of its "diagnoses." Is aggression as "real" a psychiatric symptom and "valid" a treatment target as are delusions, for example? Many in the field of emergency psychiatry argue so.

The JCAHO standards are even less clear, and unfortunately these are the standards by which most hospitals are evaluated. The tautology here rests with defining "inappropriate use."

Use of Chemical Restraint/Involuntary Medication in Psychiatric Settings

Although there is a lack of agreement on what, if anything, constitutes "chemical restraint," there is clear concern about the extent to which involuntary medication is used. This frequency is difficult to determine, since no national data exist to establish incidence rates. Chemical restraint may be used in medical settings for two basic reasons: medical and behavioral management. Medical emergency departments often restrain patients with agents such as etomidate to institute life saving procedures or diagnostic evaluations. Psychiatric emergency and inpatient services most often employ chemical restraint (typically high potency conventional antipsychotics, atypical antipsychotics, or benzodiazepines) to contain violence risk. One approximation of the frequency with which chemical restraint is used in psychiatric emergencies may be deduced from recent reports. In an article by McGaha et al. examining statewide involuntary commitment data, 3 in 1000 Florida citizens were involuntarily detained for mental health evaluation in the year 2000.[16] Data from Currier et al. suggest that 16% of patients seen in U.S. psychiatric emergency services receive involuntary medications.[17] Making a large assumption that these numbers are generalizable to the U.S. population (281,421,906 per 2000 census data), at least 135,000 persons per year receive chemical restraint in U.S. psychiatric emergency services alone. This is likely to be an underestimate, since Florida's data may be incomplete. The scope of this practice in other medical and psychiatric settings remains speculative, but is likely to be much greater.

The Debate on the Chemical Restraint Concept in the Mental Health Literature

Although it is clear that chemical restraint/involuntary medication is frequently employed, the available literature to guide clinicians is skeletal. For example, a Medline search using the keyword "chemical restraint" yielded 123 articles published between the years 1966 and December of 2002. Only 30 of these articles related to humans, the rest pertaining to veterinary practice.

Most of the literature to date focuses on the use of physical restraint and seclusion in hospital-based psychiatric settings. In an influential 1994 article in the *American Journal of Psychiatry*, Fisher reviewed the literature published since 1972.[18] *[Editor's note: for a description of a restraint and seclusion reduction program, see the article by Fisher, p. 7.]* Contents of articles were reviewed using the categories of indications and contraindications; rates of seclusion and restraint, and demographic, clinical, and environmental factors that affect these rates; effects on patients and staff; implementation and training. The author concluded that restraint and seclusion are efficacious in preventing injury and reducing agitation. However, he acknowledged that restraint and seclusion have negative physical and psychological effects on patients and staff. Other authors support the clinical necessity of some degree of coercive care in a variety of types of facilities that deal with acutely ill clients.[19, 20]

Policy Statements Concerning Chemical Restraint

In the wake of highly publicized adverse outcomes, federal regulators have focused attention on a wide variety of settings in which the use of restraint and seclusion is common. Early scrutiny of the use of chemical and physical restraint first targeted nursing home populations. Legislation in the Omnibus Reconciliation Act of 1987 and the Commission on Accreditation of Rehabilitation Facilities standards of 1993 led to a significant decrease in the use of restraint in long-term care.[21] More recently psychiatric and medical inpatient facilities have come under increased scrutiny. An October 1998 investigative series in *The Hartford Courant*[22] and a recently released

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report by the General Accounting Office, Mental Health: Improper Restraint or Seclusion Use Places People at Risk (GAO/HEHS-99-176),[23] have increased public interest in and prompted congressional concern about the use of restraint and seclusion in residential facilities. The *Hartford Courant* database is now public record and contains information about a total of 137 deaths. Based on information supplied in the series, 57 (42%) deaths occurred in residential settings, with the remainder in psychiatric or general hospitals. Children accounted for nearly 26% of these deaths. Using statistical modeling, the *Courant* authors estimated that there are between 50 and 150 deaths of all age groups each year as the result of restraint. While the methodology employed to generate this estimate is open to question, the large number of deaths cited galvanized political momentum to regulate restraint use in healthcare settings more tightly.

Until recent federal initiatives, the regulation of restraint and seclusion was haphazard. According to data from the National Association of State Mental Health Program Directors, in 1999, only 6 of 36 states responding to a survey had regulations that specifically addressed the use of restraint and seclusion in minors.[24] This report and others increased public and congressional concern about the use of restraint and seclusion in residential treatment facilities.

In August 1999, the Health Care Financing Administration (HCFA), now the Centers for Medicare and Medicaid Services (CMS), issued rules for the use of restraint and seclusion in general as well as for psychiatric hospitals that participate in Medicare and Medicaid.[14] The main tenet of the CMS regulations concerning the use of medications is that the distinction between treatment and chemical restraint hinges on whether medication is given as a part of a plan of care or is used solely to control behavior. Several of the main points of the new regulations are discussed below.

First, patients have the right to be free of restraints that are not medically necessary or are used as a means of coercion, discipline, convenience, or retaliation by staff. Restraints can only be used in the interest of a patient's well-being and only after less restrictive interventions have proven ineffective. The use of restraints must be in accordance with the order of a physician or other licensed independent practitioner and should never be written as a standing order. Finally, the restrained patient must continually be assessed, monitored, and reevaluated. The duration of restraint orders is limited to 4 hours for adults, 2 hours for children and adolescents, and 1 hour for patients under 9. During episodes of restraint, continuous in-person or combined in-person and audiovisual monitoring is recommended. All deaths that occur while a patient is in restraints or seclusion must be reported to the CMS. Further, restraints may not be used as a coercive measure for a patient refusing treatment as it is within the rights of a competent patient to refuse treatment.[14] The CMS regulations also include a 1-hour rule, which requires that a physician or licensed independent practitioner make a face-to-face assessment of a restrained or secluded patient within 1 hour of initiation. Preliminary reports suggest that the CMS rules have contributed to a reduction in the use of restraint and seclusion in the psychiatric hospital setting. In one study of general psychiatric wards serving youth, adults, and the aged, the overall number of restraint and seclusion episodes decreased by more than 50% in the 3 months after the rules went into effect, while the mean duration of restraint episodes decreased by 40.8% overall. Importantly, there was no increase in injuries to staff or others.[25] While not treated explicitly in the analyses, it is the impression of this author that physical restraint reduction depended on quicker and more aggressive pharmacological intervention before patients could reach a "crisis" threshold.

Other recent legislative and policy measures extend greater protections against the injudicious use of restraint and seclusion to other care sites. Recently, efforts have been made on a federal level to regulate and thus reduce the use of physical and chemical restraint in residential treatment facilities. The Children's Health Act of 2000 delineated that any federally funded non-medical or community-based facility was to "protect and promote the rights of each resident of the facility, including the right to be free from physical or mental abuse, or corporal punishment..."[26]

However, federal measures to regulate restraint and seclusion do not affect facilities that depend solely on state or local funding. The use of restraint and seclusion in nonmedical settings such as residential treatment facilities, juvenile correctional facilities, and jails poses additional concerns, particularly since older and younger recipients may be at elevated risk for adverse medical outcomes and medical oversight in residential treatment facilities may be inadequate. Many mental health professionals now consider jails, detention centers, and prisons as the de facto "alternate mental health system."[27] However, forced interventions, such as physical or chemical restraint, may be used very differently in these environments. Whereas such measures may be used strictly to assure client and staff safety in mental health settings, in criminal justice settings these

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interventions may be used as a form of punishment. Furthermore, time limits and the type of regulatory oversight found in healthcare settings may often be lacking in criminal justice settings, and indeed, in residential settings as well.[28] Until these issues are addressed, the quality of care in many medical, residential, and criminal justice facilities will continue to be inadequate, despite heightened federal standards.

What the Experts Said about the Chemical Restraint Concept

Nearly all of the experts surveyed for the *Expert Consensus Guidelines on Behavioral Emergencies*[11] believe that medications used to treat specific psychiatric diagnoses should be considered treatment measures rather than chemical restraints. The majority of experts further stated that drugs used to treat symptoms of psychiatric disorders could be considered treatment, even in the absence of a provisional diagnosis. Moreover, three quarters of the psychiatrists surveyed rejected the idea that any medication that is administered without a plan of treatment would be considered chemical restraint. While controversial, respondents indicated that any medication taken orally is a voluntary form of treatment, regardless of the amount of influence exerted by the staff.

Summary

Given the relative frequency with which forced medication occurs, there is very little research available to understand the use and abuse of such care. Clearly, regulatory oversight is increasing and will continue to do so in the foreseeable future. The issues I discuss in the next section concerning precipitants of emergency care and treatment selection are germane whether one sees these practices as a form of coercion or a form of patient-focused intensive care.

FACTORS THAT MAY CONTRIBUTE TO USE OF CHEMICAL RESTRAINT

Little research to date focuses on specific risk factors for receiving chemical restraint. However, it seems logical that risks for chemical and physical interventions may be similar. The literature of physical restraint is more robust, and may shed light on the differential risk of medication exposure. In a study of New York adult psychiatric wards, Way and Banks demonstrated that restraint use was independent of patient clinical or demographic attributes and was more a function of institutional factors and norms.[28] Part of these norms may include cultural stereotypes about which classes of persons require highly coercive treatment. In other work, specific patient characteristics are associated with increased risk of being physically restrained in institutional settings. West found that low intelligence, verbal comprehension, reading level, age, height, weight, length of institutionalized time, and history of abuse are predictive of higher restraint rates.[30] Level of staff experience was inversely related with restraint rates. Specific disabilities such as hearing impairment have also been associated with increased risk of the application of restraint.[31]

Institutional Characteristics

Aside from patient factors, several institutional characteristics are implicated in the excessive use of restraint. In the case of children, for example, Shaughnessy identified the following characteristics of residential care facilities that predispose to reliance on restraint: 1) organizational factors (e.g., poor staff-administration relations, little accountability in large institutions, lack of cooperation between departments, policies that require specialized staff, heavy paperwork duties for therapists), 2) incompetent staff, 3) staff turnover, 4) a remote location that isolates children from parents, 5) inappropriate placement of children as a stopgap measure, 6) the use of medication as a form of social control, 7) premature discharge, and (8) inadequate follow-up.[32] Blatt also described a correlation between stressful working conditions and subsequent coercive treatment patterns.[33]

Staff Perceptions and Attitudes

Factors related to abusive practices include not only patient characteristics and institutional culture and working conditions, but also individual staff perceptions and attitudes. In a survey of 100 direct care providers in residential treatment centers, Rindfleish and Baros-Van Hull found that willingness to use force was associated with management of routines within the organization, degree of participation in decision making, size of the community in which the caregiver was raised, age of the caregiver, and degree of resentment towards children.[34] Numerous authors have argued that staff training with an emphasis on cultural competency can significantly lower the use of restraint and seclusion without adversely impacting patient or staff safety. Goren et al.

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found the use of restraint and seclusion was reduced in children's psychiatric hospitals without increasing medication use through education, improved staff communication, policy revision, and engagement of families in care processes.[35] Staff training has also proven useful in reducing restraint use in psychiatric inpatient settings serving adults[36] and children and adolescents[37] and in residential settings serving developmentally delayed youths.[38]

Staff Safety

No data exist on the number of injuries to staff members that occur during behavior management interventions in healthcare or community settings. However, recent data from Currier et al. documented an average staff assault rate in psychiatric emergency services in excess of 6 events per year, more than 50% of which result in lost time from work.[17]

Violence towards staff is likely to be underreported due to the belief of healthcare providers that assault risk is part of the job. Underreporting may also be related to cumbersome reporting policies, perceptions that reporting is ineffectual, or employee fears that assaults may be perceived as a result of poor job performance. Whatever the reason for underreporting, it is clear that, in many settings, caring for disturbed clients is a dangerous endeavor, and that reliance on coercive care must be interpreted in that context.

RECOMMENDATIONS IN THE *EXPERT CONSENSUS GUIDELINES*

The *Expert Consensus Guidelines on the Treatment of Behavioral Emergencies*[11] present recommendations for the use of physical and chemical restraint and seclusion. In the following discussion, I review the recommendations concerning use of emergency medications (whether considered chemical restraint or not) derived from the *Guidelines* and discusses treatment developments since the survey on which the guidelines were based was conducted in 2000.

The expert panel largely agreed that the use of restraint is usually appropriate for patients who pose an acute danger to other patients, staff, or themselves. Restraint may sometimes be appropriate to prevent an involuntary patient from leaving prior to assessment or transfer to a locked facility. The most important decision in managing an acutely agitated patient is determining when a person's behavior and lack of compliance with treatment require emergency intervention.

According to the *Expert Consensus Guidelines*, emergency intervention is always necessary when a patient is directly threatening or assaultive, and is usually appropriate for a patient with a constellation of symptoms such as refusal to cooperate and heightened motor restlessness. In earlier years, justifications for restraint use included prevention of property damage, protection of ward milieu from socially unacceptable behaviors, decreasing stimulation, and as a contingency in the behavior therapy of dangerous behaviors. Some of these reasons, particularly using restraint as a contingency in a behavioral therapy plan, are no longer deemed acceptable.[18] Still, the decision to initiate emergency interventions remains difficult, particularly for inexperienced clinicians, as there are no agreed upon measures of agitation.

Assessment

In the event of a behavioral emergency, prompt control of acute aggression is essential to ensure patient safety and to facilitate a more typical patient-physician relationship. An important component of the initial evaluation of a patient in a behavioral emergency involves investigation of potential medical etiologies for the presentation. Psychiatric personnel performing an initial medical evaluation should take vital signs, obtain a medical history, and perform a visual examination. Ideally, the most thorough assessment possible should be performed before initiating emergency treatment. In actual clinical practice, complete assessment is rarely achievable. According to the *Expert Consensus Guidelines*, in urgent situations it is sufficient to perform a brief psychiatric assessment prior to initiating a plan of care.

Least Invasive Intervention

When selecting an emergency intervention, it is advisable to begin with the least aggressive approaches such as verbal interaction, offering food or beverages, or giving the option of voluntary medication. If these methods do not ameliorate the situation, a show of force is appropriate. If all less invasive measures fail, emergency medication, physical restraint, or seclusion may be appropriate. The *Expert Consensus Guidelines* indicated that even under adverse conditions, it is important to keep the desires of patients in mind. While patient satisfaction may not be the primary concern in a behavioral emergency, such issues impact the success of treatment and compliance with follow-up.

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To ascertain treatment preferences and determine which measures were viewed as least disturbing, Sheline and Nelson anonymously surveyed clients in a public psychiatric emergency service.[39] Among the patients surveyed, 64% preferred medication while 36% favored physical restraint. The patients considered the most favorable management strategies to be benzodiazepines (31%), neuroleptics (26%), seclusion (24%), and restraint (10%).[39]

In practice, physical and chemical restraint are commonly used in tandem, and it is increasingly rare for physical restraint to be used in the absence of some type of sedative agent. Diaz stated, "Any maximal physical restraint without chemical restraint is at all times unacceptable."[40] Medication can be useful in reducing the amount of time a person spends in restraints.[40]

Another study examined consecutively admitted, forcibly medicated, English-speaking, acute care inpatients after their discharge.[41] Among the 30 patients studied, 47% received forced injections, while the remainder received oral medication under duress. At post-discharge interviews, the following were the most commonly reported experiences by patients: fear of side effects (57%), belief that nothing was wrong with them (30%), fear of being weakened by the medication (20%), and feeling that the medication was poison (20%). Surprisingly, 60% of patients retrospectively agreed with the decision to use involuntary treatment and 53% reported that they were more likely to comply with medications in the future.

The Use of Medications for Emergency Intervention

As noted above, in most cases medications are used in conjunction with physical restraint. If a patient becomes calmer while in restraints, the experts were divided as to whether or not to use oral medications or if medication should be avoided. However, the expert consensus indicated that if a patient remained agitated and violent while in restraints, parenteral medications should be administered.

The paramount concern in prescribing a medication for an agitated patient is the effectiveness of the drug. At the time of this survey, haloperidol, droperidol, oral atypical antipsychotics, and lorazepam were recommended drugs for managing aggressiveness. Droperidol was favored due to its highly sedative properties and rapid speed of onset in intramuscular form.[42, 43] Since that time, droperidol has been withdrawn from the European market[44] and has received a "black box" warning in the United States due to concerns about its potential to cause QT prolongation.[45]

Comparisons of lorazepam 2 mg with haloperidol 5 mg suggest that lorazepam is superior on measures of aggression and clinical global improvement.[46, 47] However, a comparison of droperidol and lorazepam in violently agitated patients requiring restraint indicated that droperidol was associated with more rapid and effective sedation at 10, 15 , 30, and 60 minutes.[42]

One of the most common medication strategies for managing psychotic agitation is to use a combination of haloperidol 5 mg and lorazepam 2 mg. For a review of this approach, see Allen.[48] While effective in the short-term with or without benzodiazepines or anticholinergic drugs, high-potency butyrophenones may have unappreciated adverse consequences, including persisting risk of extrapyramidal symptoms. This standard combination may be supplanted in the near future as atypical agents are increasingly used in the emergency setting.

Atypical antipsychotics were slow to gain popularity in emergency situations due to slow titration schedules and, until recently, the lack of intramuscular preparations. The prototype atypical antipsychotic, clozapine, has proven efficacious in treating symptoms of agitation. In a pre/post design, Mallya et al. demonstrated that treatment with clozapine resulted in fewer instances and shorter duration of restraint/seclusion compared with conventional antipsychotics.[49] However, clozapine is not indicated for rapid tranquilization because of its extensive side-effect burden.

Beasley et al. examined the incidence of aggression among patients with DSM-III-R diagnoses of schizophrenia, schizophreniform disorder, or schizoaffective disorder who were treated with olanzapine (n = 2284), haloperidol (n = 810), or placebo (n = 118).[50] Patients were assessed based on spontaneously reported adverse event data and clinician-assessed rating scale items on the Brief Psychiatric Rating Scale (BPRS).[51] Analyses of BPRS hostility items indicated that patients treated with olanzapine displayed statistically significant decreases in mean hostility scores compared with patients treated with placebo and comparable scores to patients treated with haloperidol.[50]

A recent study found that a combination of risperidone plus lorazepam was as efficacious as a combination of haloperidol plus lorazepam for the treatment of psychotic agitation.[52] Currier and Simpson studied 60 psychotic patients admitted to a large psychiatric emergency service who required emergency medication for violence and agitation. Patients were offered a combination of risperidone, 2 mg liquid concentrate, and

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oral lorazepam, 2 mg, as an alternative to the standard combination of haloperidol, 5 mg IM and lorazepam, 2 mg IM. Both groups showed a significant improvement in five agitation subscales of the Positive and Negative Symptoms Scales (PANSS),[53] in scores on the Clinical Global Impression (CGI) scale,[54] and in time to sedation. One patient treated with haloperidol experienced acute dystonia, while no patients receiving risperidone displayed significant side effects.[52] Czobar et al. compared the use of risperidone and haloperidol for the treatment of hostility in a multicenter, placebo-controlled, double-blind study.[55] Hostility was measured by the hostility item on the PANSS. Risperidone was found to have a greater selective effect on hostility than haloperidol or placebo, and the improvement in hostility associated with risperidone did not appear to be dose-related.[55]

Quetiapine has received some consideration as a potentially useful therapeutic option for the treatment of psychotic agitation.[56] One clinical trial compared five fixed doses of quetiapine with both haloperidol and placebo for the treatment of aggression and hostility. Patients were assessed using the BPRS Factor V score, which includes hostility, excitement, suspiciousness, and uncooperativeness. A specific beneficial effect on measures of hostility and aggression was found in all groups treated with quetiapine and this effect reached statistical significance at a dose of 150 mg. This effect did not occur with haloperidol. Responder rates, as determined by a > 40% reduction on the BPRS, indicated that a significantly greater proportion of patients receiving quetiapine responded over the dose range of 150 mg–750 mg compared with those receiving placebo. Improved outcomes may be related to limited extrapyramidal symptoms versus high potency conventional antipsychotics.[56]

Intramuscular ziprasidone became available in the United States in the fall of 2002. One 7-day, randomized, open-label, multicenter trial examined the efficacy and tolerability of intramuscular ziprasidone and intramuscular haloperidol and the transition from intramuscular to oral treatment in hospitalized patients with acute psychotic agitation.[57] Flexible intramuscular doses of ziprasidone (n = 90) or haloperidol (n = 42) were given for 3 days followed by oral treatment up to day 7. Ziprasidone was initially given at a dose of 10 mg IM with subsequent doses of 5–20 mg given every 4–6 hours as needed (maximum daily dose 40 mg/day). Oral doses of 80–200 mg/day ziprasidone were given subsequently. Patients in the haloperidol group received an initial IM dose of 2.5–10 mg, followed by subsequent doses of 2.5–10 mg every 4–6 hours as needed, and were then switched to oral haloperidol. Ziprasidone produced greater decreases in total BPRS, BPRS agitation items, and CGI Severity Scales compared with haloperidol and was associated with a lower incidence of movement disorders and a reduced need for anticholinergic medications.[57]

Lesem et al. also evaluated the efficacy and tolerability of rapid acting IM ziprasidone in the treatment of patients with psychosis and acute agitation.[58] In this 24-hour fixed-dose, double-blind clinical trial, patients were randomly assigned to receive up to 4 injections (every 2 hours prn) of 2 mg (n = 54) or 10 mg (n = 63) IM ziprasidone. As indicated by the Behavioral Activity Rating Scale, 10 mg ziprasidone rapidly reduced symptoms of acute agitation and was more effective than a 2 mg dose up to 4 hours after the first IM injection. No acute dystonia or behavioral disinhibition was observed, although one patient did experience akathisia at a dose of 10 mg. Ziprasidone 10 mg IM was judged to be rapidly effective and well-tolerated in short-term management of agitated psychotic patients.[58]

Jones et al. reviewed the results of three clinical trials of IM olanzapine (which is currently in late stage development) and examined positive symptom efficacy as judged by the BPRS agitation subscale.[59] In one open-label study, patients were treated with 2–10 mg of intramuscular olanzapine, 1–4 times daily for up to 3 days. This was followed by oral administration of 10–20 mg/day olanzapine for 2 days. The patients displayed a mean decrease from baseline in the BPRS positive subscale score. Jones et al. also described a randomized double-blind study in which 311 acutely agitated patients were given up to 3 injections of IM olanzapine (7.5 mg/injection), IM haloperidol (10 mg/injection), or IM placebo. Following a 24 hour period in which medications were given intramuscularly, there was a 4-day period in which oral olanzapine was given at a dose of 5–20 mg/day to those patients who had previously received IM olanzapine or placebo, and haloperidol was given at a dose of 5–20 mg/day to those patients who had previously received IM haloperidol. This study indicated a statistically significant decrease in the BPRS positive subscale in patients treated with either olanzapine or haloperidol but not placebo.[59] A more recent report of a similar study provides similar findings, and indicates that intramuscular olanzapine was better tolerated than haloperidol, with less treatment-emergent parkinsonism.[60] This study further suggests that 10 mg is the most effective dose for the majority of patients.

Recent findings need to be considered in the context of the inherent difficulty of conducting studies in agitated, potentially psychotic individuals. Since, with few exceptions, these studies of atypical agents correctly required

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informed consent, the levels of agitation of study participants was well below that generally seen in clinical populations. Significant medical and substance abuse comorbidities were also screened out, making comparisons with emergency room populations less clear. Nonetheless, as clinical and Phase IV data accumulate, information concerning the safety and efficacy of new atypical preparations in highly agitated "real world" patients will be easier to interpret.

Selecting an Intervention Based on Etiology

The *Expert Consensus Guidelines on Behavioral Emergencies* provide recommendations for treating behavioral emergencies attributed to various etiologies. Several more recent reviews consider various drug strategies for treatment of psychotic agitation in detail.[48, 61, 62]

One of the most critical determinations to be made is whether or not an acute agitated state can be attributed to a general medical etiology. Essential aspects of examination include taking vital signs, obtaining a medical/psychiatric history from family or other sources, talking to the patient (if possible), performing a visual examination, consultation from the medical emergency department, pulse oximetry, blood glucose, and toxicology screening. If a general medical etiology is suspected, but behavioral disturbances require immediate intervention, considerations should include physical restraint, oral medications, and a focused physical examination. When medications are indicated, the *Expert Consensus Guidelines* favor a conventional antipsychotic, a benzodiazepine, or a combination of the two. At the time of the survey, the experts suggested that delirium due to a general medical condition should be treated with a high potency antipsychotic.[11] However, several recent case reports suggest that atypical antipsychotics may be safe and effective in treating delirium, although more definitive head-to-head trials versus haloperidol need to be conducted.[63–66]

A slightly different approach should be taken if a patient's presentation is believed to be associated with substance intoxication. The initial intervention should include vital signs, talking to the patient, gathering a medical history from other sources, performing a toxicology screen, and performing a visual examination of the patient. Other recommendations include oral medications, a limited physical examination, and blood or breath testing for alcohol. As with all patients, restraint should be avoided if possible. If patients are restrained and left to "dry out," frequent safety checks are mandatory, since patients may not be able to signal for help. Clinicians should be mindful of the risk of aspiration. For alcohol-or stimulant-intoxicated patients without a pre-existing psychotic diagnosis but who require sedation, benzodiazepines are recommended. Patients intoxicated with alcohol may benefit from benzodiazepines as a prophylactic measure for seizures.

If a patient presents with agitation that is believed to be the result of a primary psychiatric disturbance, immediate intervention may be required to ensure the safety of the patient and staff. Initially, vital signs should be taken, verbal interventions should be attempted, and information should be gathered from collateral contacts. A visual examination and toxicology screen may be beneficial. Parenteral or oral medications should be administered according to guidelines specific to the provisional diagnosis.[11]

If a patient is a good candidate for oral medication, this intervention may be preferable to using IM drugs. For psychotic patients, a benzodiazepine in conjunction with a conventional or atypical antipsychotic is considered the first-line treatment. The *Expert Consensus Guidelines* series did not agree on a first line treatment for diagnoses of psychotic depression or personality disorders, although high second-line recommendations included a benzodiazepine used alone or in combination with a conventional or atypical antipsychotic agent. For a provisional diagnosis of posttraumatic stress disorder, the first-line recommendation was monotherapy with a benzodiazepine.

For some patients, oral medications are not a realistic treatment option and a parenteral preparation is necessary. For patients with a provisional diagnoses of schizophrenia, mania, or psychotic depression, combined treatment with a benzodiazepine and a conventional antipsychotic was still regarded as first-line treatment when an injectable medication was to be used. In many settings, parenteral ziprasidone has already supplanted intramuscular butyrophenones. For schizophrenia and mania, the high second-line treatment alternative was a conventional antipsychotic used alone. For psychotic depression, the high second-line recommendation was monotherapy with a benzodiazepine. There was no first-line recommendation for parenteral treatment of patients with a personality disorder alone, though the high second-line treatment was a benzodiazepine used alone or in combination with a conventional antipsychotic agent. If no data are available to explain an agitated presentation and oral medication is to be administered, the first-line recommendation is a benzodiazepine alone. As an alternative, a benzodiazepine

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may be used in combination with a conventional or atypical antipsychotic agent. There is no first-line recommendation if a parenteral preparation is required. Benzodiazepines alone or in combination with a conventional antipsychotic were suggested.[11]

In some cases patients may respond inadequately to the initial drug administration. If a patient was initially treated with a benzodiazepine or antipsychotic agent alone and no adequate response is seen after 45–60 minutes, combination treatment should be implemented or a second dose should be given. Medications should be changed if two doses prove to be totally ineffective or three or four doses are partially ineffective. At this point, reconsideration of the diagnosis may be warranted, with the possibility of delirium considered.

Safety and tolerability are always a special concern when antipsychotic agents are used in vulnerable patient groups such as pregnant women and children. Based on years of accumulated clinical experience, but very thin literature, conventional antipsychotic agents are still recommended for pregnant women. There was no consensus on the appropriate treatment for agitated children, although low-dose benzodiazepines or antihistamines were regarded as high second-line options. Studies of atypical agents in these patient groups, while difficult to carry out, will be critical.

There are certain situations in which specific agents should be avoided. Conventional antipsychotics should be avoided in patients with a significant history of extrapyramidal symptoms. Many clinicians are reluctant to use benzodiazepines in patients with a history of substance abuse or dependence because of their potential for abuse. Benzodiazepines should, however, be used for patients experiencing alcohol withdrawal. Benzodiazepines may be inappropriate for delirious individuals.

One further concern is cost of treatment. Table 1 presents the per dose cost at one academic medical center of a variety of medications that may be potentially useful in the treatment of behavioral emergencies. These costs vary across institutions and will likely change over time. However, as formulary restrictions become more common, clinicians will need to assess incremental clinical benefits of newer medications in light of the markedly increased costs associated with these agents.

Table 1. Cost of drugs used to treat agitation, University of Rochester, December 2002

Drug	*Route*	*Dose*	*Cost (U.S. $)*
Diphenhydramine	IM	50 mg	0.90
Haloperidol	PO tablet	5 mg	0.03
	PO liquid	5 mg	0.62
	IM	5 mg	2.94
Lorazepam	PO tablet	2 mg	0.46
	IM	2 mg	0.98
Olanzapine	PO tablet	10 mg	7.60
	PO quick dissolve	10 mg	2.10
Quetiapine	PO tablet	100 mg	2.10
Risperidone	PO tablet	2 mg	3.89
	PO liquid	2 mg	5.77
Ziprasidone	IM	20 mg	35.00

CONCLUSIONS

Practitioners working in acute care psychiatry are charged with containing volatile and aggressive behavior while protecting patients' rights and avoiding undue coercion, a balance that is not easy to maintain. Recently, public outcry concerning perceived abuse of restraint authority has promoted vastly increased regulatory oversight. In many healthcare settings, emergency medications are viewed as a more humane alternative to physical restraint. However, Federal language appears to equate chemical and physical restraint, and characterizes both as extraordinary practices not to be undertaken lightly. Practitioners now have access to a variety of newer agents in oral and parenteral forms, including atypical antipsychotic agents. Based on available literature, these drugs appear to be as effective as older medications, but may avoid adverse side effects and prove more acceptable to patients. Findings from the *Expert Consensus Guidelines* suggest that earlier intervention with appropriate treatment can avoid behavioral emergencies, and should not be characterized as chemical restraint in the context of a symptom-based, but not necessarily a disorder-based, plan of care.

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